

ReelCast

Project Proposal

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Chapter 1: Introduction and Background

1.1 Context

In the current era, Social Commerce is growing rapidly. The primary drivers of this ecosystem include online merchants, brand owners, and influencers (such as content creators and affiliate marketers) who need to distribute promotional content consistently to engage with their target customer groups and drive sales. To capture audience attention effectively, entrepreneurs and influencers rely heavily on short vertical videos (Reels) on platforms like Instagram and Facebook to advertise products and reach target customers. Reels are short-form, vertical videos (typically under 60 seconds) that have become an indispensable marketing tool because their algorithm-driven feeds offer immense organic reach, allowing brands to showcase products dynamically and seamlessly integrate purchasing options.

Insights gathered from recent user interviews confirm this heavy reliance on short-form videos but also highlight significant operational hurdles. For example, an interview with Tan, a university student managing an online custom screen-printed t-shirt business, revealed that balancing a demanding schedule leaves very little time for marketing. Although he attempts to create "shopping basket" style affiliate clips to drive sales, his posting schedule remains highly inconsistent. Tan identified video editing as his primary hurdle, noting that a lack of prior experience creates a steep learning curve. This often leads him to skip editing altogether, limiting both the quality and frequency of his promotional videos.

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Figure 1: An example of short-form vertical videos (Reels) on social media platforms featuring embedded product links.

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1.2 Core Problems

To illustrate the current bottlenecks in the market, consider the typical scenario of an online merchant launching a new product campaign. First, they must record raw footage, transfer it to complex video editing software, and spend hours cutting, adding captions, and overlaying graphics to make the clip engaging. Once the video is finally rendered, they must manually upload it to Instagram, Facebook, and TikTok individually. Furthermore, for each platform, they have to manually generate and paste specific affiliate tracking links into the comments or descriptions.

Despite the high potential of Reels, creators face significant bottlenecks from this repetitive, multi-step workflow:

- **High-Barrier Production:** Creating high-quality commercial videos requires advanced editing skills and takes hours to complete.
- **Fragmented Tools (The "Silo" Effect):** Creators are forced to juggle multiple disjointed applications. They must generate a video in one app, manually transfer it to a scheduling tool, and separately manage their affiliate links in another.
- **Complex Link Management:** Embedding and tracking affiliate links across different platforms is a repetitive and error-prone manual task. Furthermore, navigating strict API limitations and policies across various social media networks adds significant administrative overhead. This fragmented workflow creates unnecessary friction and severely slows down content creation.

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1.3 System Overview

To address the profound limitations and workflow frictions currently faced by creators and merchants, this project introduces **"ReelCast"—an all-in-one, automated web application specifically designed to empower users to effortlessly create, manage, and distribute premium short-form video content.**

Rather than relying on a fragmented toolset, ReelCast acts as a centralized hub that seamlessly bridges the gap between content creation and e-commerce. As noted in the user interview, target users expressed strong enthusiasm for the ReelCast concept. They specifically highlighted that having a unified system for automated product video generation and multi-platform distribution would completely eliminate their technical barriers, save hours of frustration, and effortlessly open up new market opportunities for their businesses without adding to their workload.

1.4 Proposed Solution: ReelCast

To turn the ReelCast concept into reality, the system integrates advanced Generative AI (Google Veo and Gemini) to reduce video production time from hours to minutes while maintaining high commercial quality. The proposed solution encompasses the following core features:

- **AI-Powered Reel Generation:** Automatically generates marketing-focused B-roll videos and platform-specific captions based on product inputs, utilizing pre-defined prompt templates to ensure the content is highly optimized for commercial success.

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- **Comprehensive Product Library:** Allows users to manage inventory organized by specific marketing campaigns or projects, embed smart product links, and track promotional performance in a centralized hub.
- **Preview & Approval Workflow:** Ensures users maintain **full creative control**—specifically the ability to review AI-generated B-roll, edit Gemini-crafted captions and hashtags, adjust product image overlays, and regenerate media if the initial AI outputs are unpredictable— before any content is published.
- **Automated Distribution:** Seamlessly publishes and schedules the finalized media directly to target social media platforms. To ensure a robust Proof of Concept (PoC) and effectively manage API constraints, the initial Minimum Viable Product (MVP) will focus on full integration with two primary platforms: **YouTube Shorts and TikTok**. However, to navigate strict API policies and avoid unexpected rate limits, the TikTok integration during the MVP phase will be developed and tested primarily within a Sandbox Environment.

1.5 Expected Impact

By eliminating the friction between content creation and commerce, this tool allows online merchants, affiliate marketers, and brand owners to scale their marketing efforts and focus strictly on growing their business.

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Chapter 2 Literature Review

2.1 Business Review

2.1.1 Overview

In the highly competitive landscape of Social Commerce, businesses must adopt an effective Digital Marketing Strategy to survive and scale. While traditional marketing relies on static images and text-based posts, modern strategies have shifted heavily toward dynamic, algorithm-driven formats. Among the various methods available, short-form vertical videos (collectively referred to as "Reels" across platforms like Instagram, Facebook, and TikTok) have proven to be the most engaging and lucrative approach. This is supported by market data indicating that short videos generate significantly higher engagement and conversion rates compared to any other media format.

Understanding Reels and the Traditional Workflow: Reels are essentially bite-sized, full-screen video content, typically under 60 seconds, designed to quickly capture user attention and deliver a concise message or product showcase. From a business operational perspective, the traditional workflow for producing and distributing a commercial Reel is highly demanding. It generally involves the following sequential steps:

- 1. **Conceptualization & Filming:** Planning the product showcase and recording raw footage.
- 2. **Post-Production (Editing):** Using external software to cut clips, add trendy audio, insert transitions, and overlay text.

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3. **Platform-Specific Formatting:** Adjusting the final video to meet the distinct technical requirements of each social media platform.
4. **Manual Distribution & Link Management:** Individually uploading the video to each app and manually embedding the correct affiliate tracking links in the captions or comments.

The **"ReelCast"** project focuses on building a sophisticated, AI-driven web application to streamline this entire digital marketing workflow. By combining advanced Generative AI models from Google Vertex AI (Veo and Gemini) integrated with pre-defined marketing prompt templates, the system offers a complete solution. It streamlines everything from campaign-based product input and preview/approval to monetized content delivery and post scheduling.

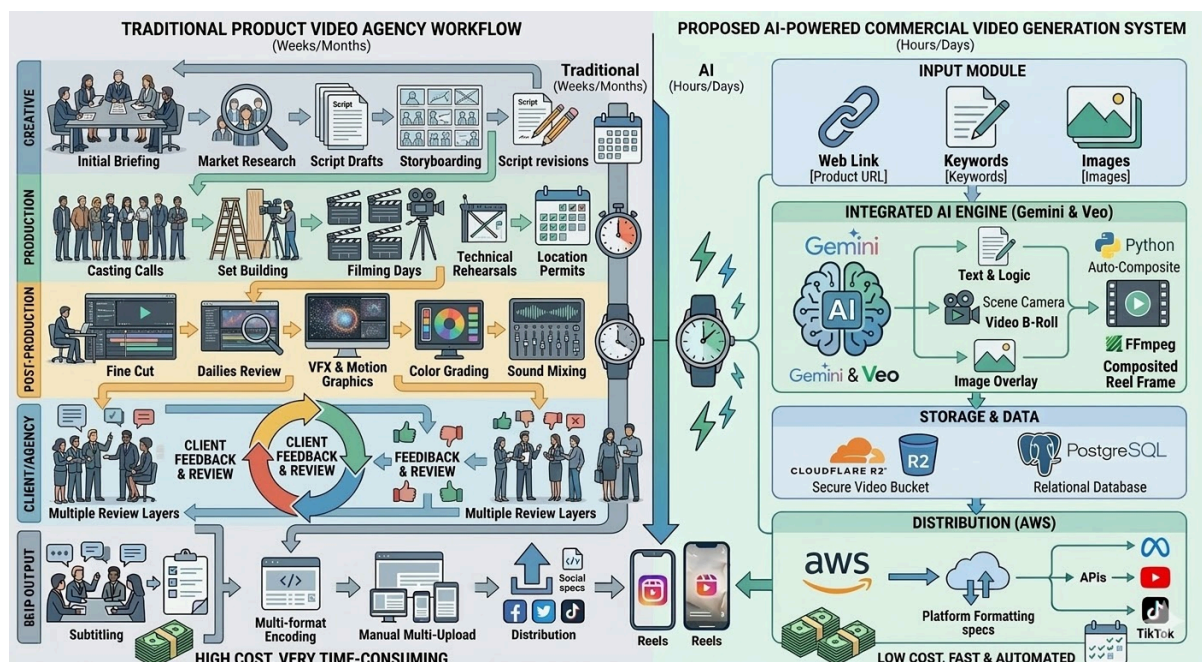


Figure 2: A comprehensive comparison between the traditional, time-consuming video production workflow and the automated, AI-powered ReelCast system.

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2.1.2 Core Business Problem & Opportunity

- **The Problem:** Online businesses struggle to consistently execute their digital marketing strategies due to the high production costs (writing, filming, editing) associated with the traditional Reels workflow. Additionally, managing vast product inventories, tracking the performance of scattered affiliate links, and navigating the operational burden of manually posting to each platform creates significant administrative overhead that stifles business growth.
- **The Opportunity:** Leverage the latest-generation AI to automate content production and management. Generative AI (like Google Veo for video and Gemini for text) can execute the post-production and formatting steps almost instantly. Coupled with a campaign-centric Product Library, businesses can seamlessly generate marketing-focused content, retain creative control through a preview workflow, and track promotional success in one unified ecosystem, completely bypassing the traditional workflow bottlenecks.

2.1.3 Target Market

- **Online Merchants:** Direct-to-consumer businesses needing to scale their digital presence.
- **Affiliate Marketers:** Individuals or agencies focused on driving traffic and earning commissions, who manage large numbers of products and links.
- **Brand Owners:** Businesses managing extensive product catalogs that require a centralized system to link inventory with marketing assets and track overall promotional performance.

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2.1.4 Market Trends & Industry Landscape

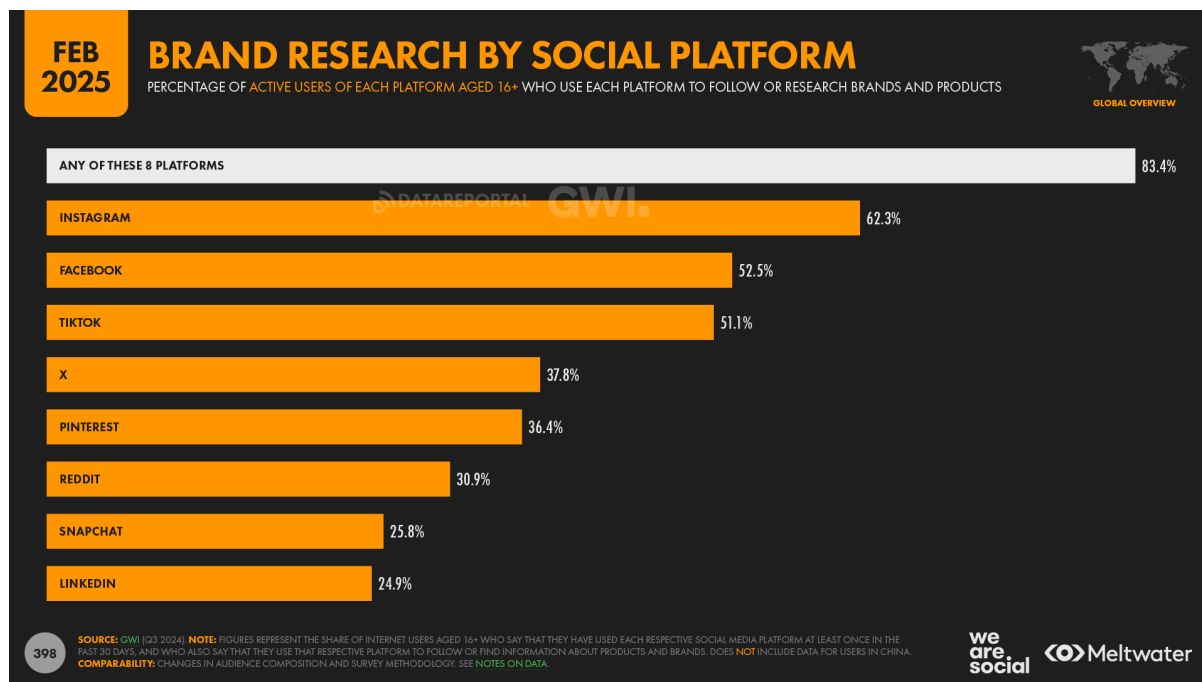


Figure 3: Percentage of active social media users worldwide who use platforms to follow or research brands and products (Source: Digital 2025: the state of social media in 2025).

- Social Commerce & Platform Dominance:** The behavior of purchasing through social media (Social Commerce) has grown significantly, particularly in using platforms to follow brands and research products. According to the *Digital 2025 Global Overview Report* (as shown in Figure 2.x), **Instagram (62.3%), Facebook (52.5%), and TikTok (51.1%)** rank as the top three platforms worldwide used by active users to research brands and products. This statistic strongly aligns with the core objective of the ReelCast system, which focuses on generating and distributing short-form videos to these specific platforms to maximize sales opportunities for users.

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- **Short-form Video Growth:** Short-form video remains the fastest-growing content format. Currently, short vertical videos generate significantly higher engagement rates compared to long-form content. This shift in consumer behavior makes platforms like TikTok and Instagram Reels the most effective channels for digital advertising.
- **Affiliate Economy:** The global affiliate marketing industry is expanding rapidly, with a vast majority of brands adopting affiliate programs to drive sales. Consumer behavior shows a strong tendency to purchase goods through short video reviews, proving that seamless integration of product links within video content is highly lucrative.
- **AI Adoption:** The integration of Generative AI in the business sector has become a new standard. A large percentage of marketers have already adopted AI tools for content creation, significantly reducing production costs and accelerating the speed of media delivery.

2.2 Business Tools/Software Review

2.2.1 Direct Competitive

There are a few tools available for automated content creation and social media distribution. Among those related to our proposed system, ReelCast, Predis.ai and InVideo web applications are the most relevant. They both have similar functions to the proposed system, such as AI-driven video generation and social media management. However, both are still different because they do not fully integrate advanced generative models like Google Veo alongside an automated affiliate link management module, a campaign-based product library, and pre-defined marketing prompt templates. The details are explained below:

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2.2.1.1 Predis.ai

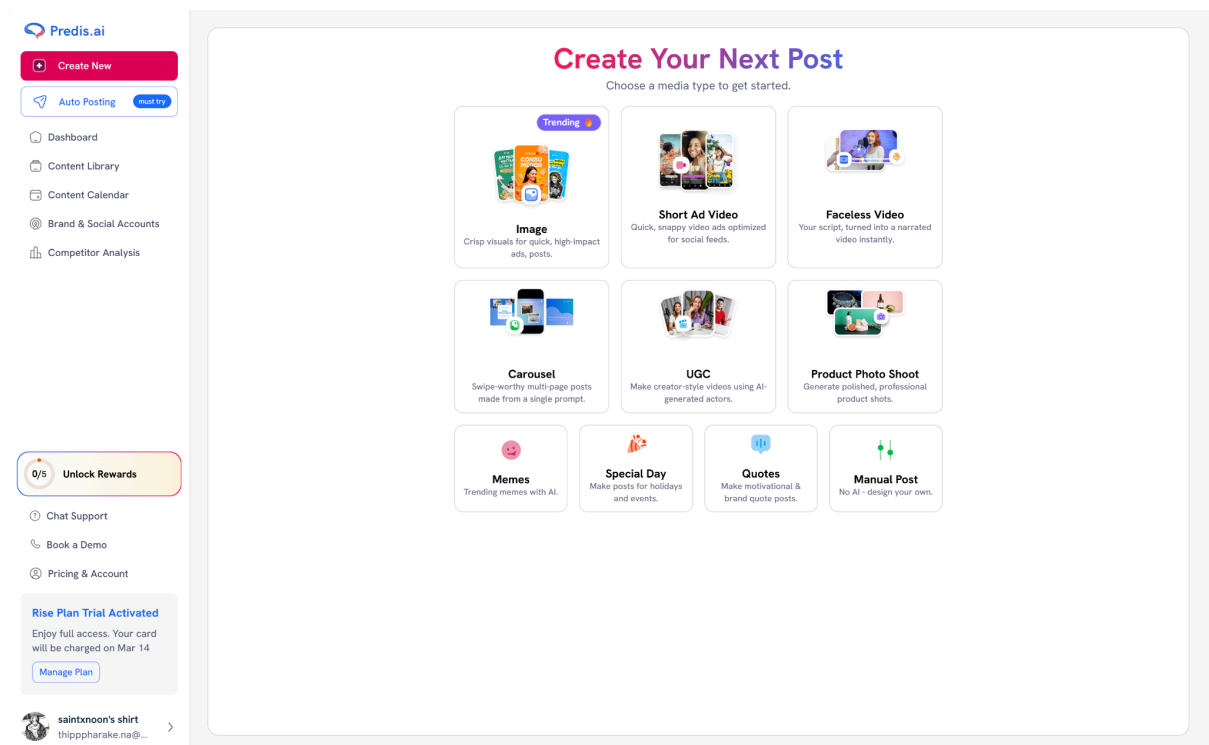


Figure 4: User interface of Predis.ai, demonstrating its AI content generation and social media scheduling dashboard.

Predis.ai (<https://predis.ai/>) is a web application that enables the user to generate social media posts, including short-form videos and carousels, using AI by inputting a simple text prompt or e-commerce product URL. After generating the content, the system can schedule and publish it directly to integrated platforms like Instagram, Facebook, and TikTok. Compared with ReelCast, this is the closest competitor. However, there are significant differences in the underlying AI technology, the lack of a dedicated campaign-centric affiliate management system, and the absence of a structured preview and approval workflow.

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- Advantages:
 - Predis.ai provides an all-in-one system for both AI content generation and social media scheduling.
 - Predis.ai supports direct API integrations with multiple major social media platforms.
 - Predis.ai allows users to generate content directly from product URLs.

- Disadvantages:
 - Predis.ai does not utilize Google Veo integrated with pre-defined marketing prompt templates for high-fidelity, commercial-focused video generation.
 - Predis.ai does not provide an automated, campaign-based product library and affiliate link management system suitable for the generated Reels.
 - Predis.ai lacks a manual upload alternative workflow, which is crucial to bypass AI-generated content penalties on platforms like TikTok.

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2.2.1.2 InVideo.io AI

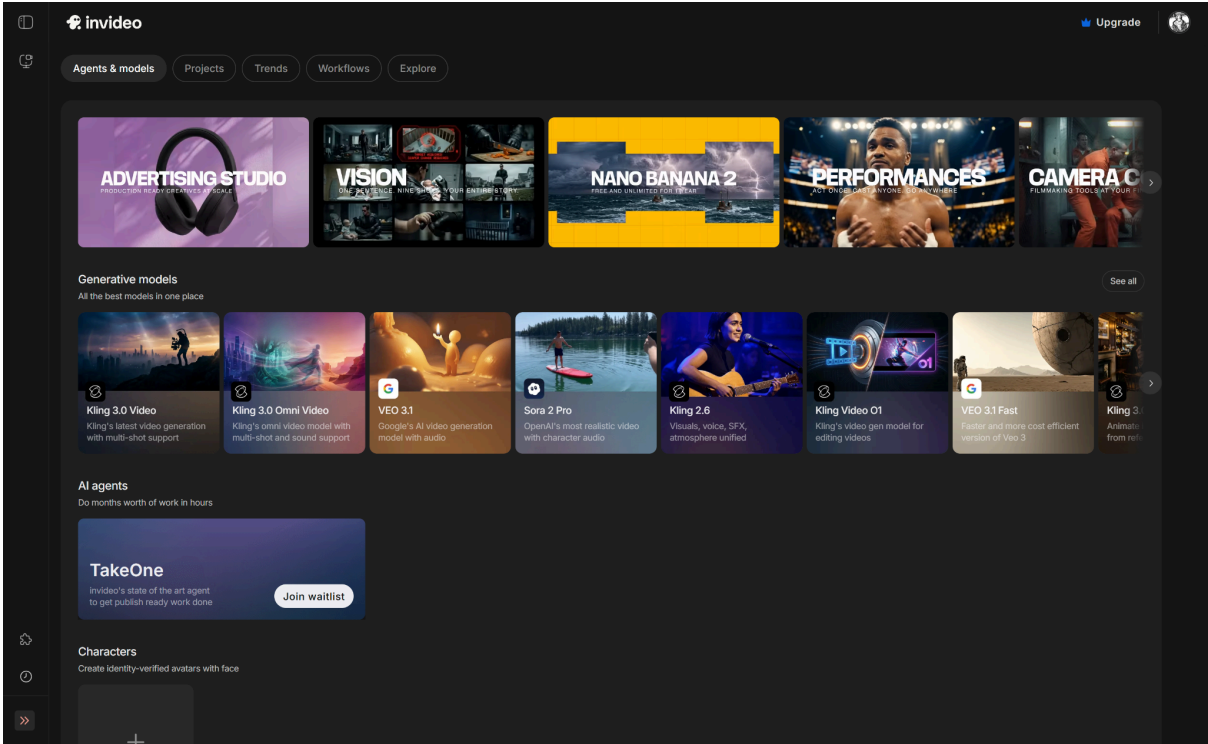


Figure 5: User interface of InVideo AI, showcasing its text-to-video generation and script-editing workspace.

InVideo AI (<https://invideo.io/>) is a web application that enables users to create publish-ready videos entirely from text prompts. It uses AI to generate scripts, select relevant stock media, and apply natural-sounding voiceovers automatically. Compared with the proposed system, this is a strong competitor regarding the AI generation module. However, there are still many differences, such as it does not have any built-in distribution module for multi-posting and it is not tailored specifically for affiliate marketers.

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- Advantages:
 - InVideo AI provides high-quality text-to-video generation capabilities with advanced scripting.
 - InVideo AI provides a vast integrated library of stock media and voiceover options.
- Disadvantages:
 - InVideo AI does not support a multi-platform distribution and scheduling module for direct automated posting.
 - InVideo AI focuses strictly on media creation and does not support project-based product tracking or affiliate link integration.
 - InVideo AI does not natively use Google Cloud Vertex AI (Veo and Gemini) for its core processing.

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2.2.2 Indirect Competitive

2.2.2.1 Sora AI (ChatGPT)

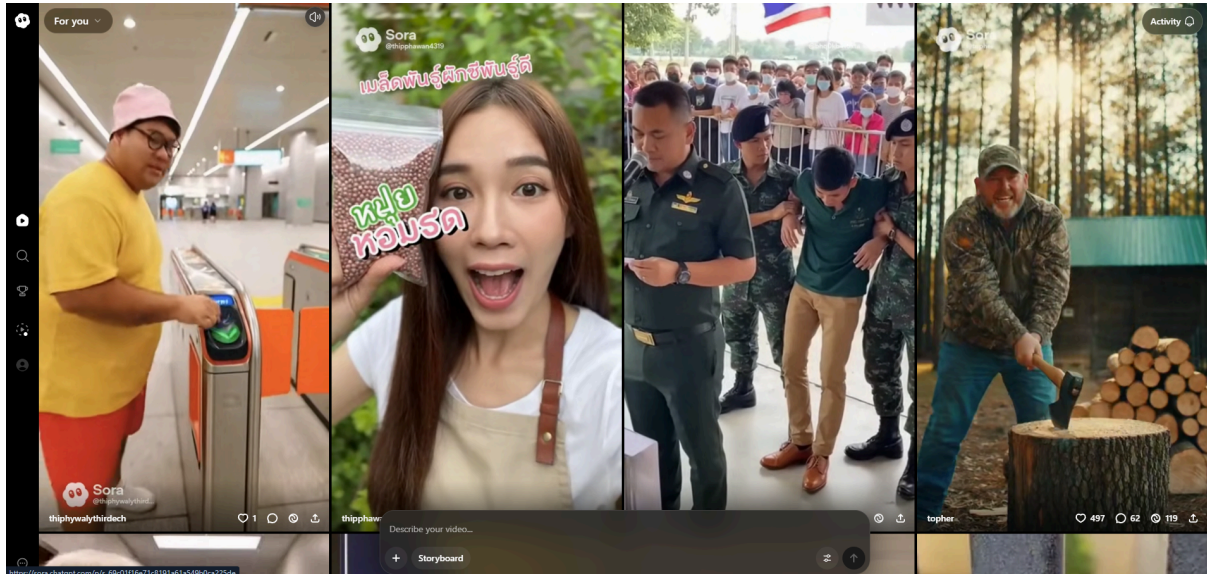


Figure 6: User interface of Sora AI, demonstrating its advanced text-to-video generation capabilities.

Sora AI (<https://sora.chatgpt.com/>) is a highly advanced generative AI model developed by OpenAI, capable of creating realistic and imaginative video scenes from text instructions. While it excels at raw video generation, compared to ReelCast, it functions strictly as a standalone generative tool rather than a comprehensive business platform. It lacks an integrated environment for e-commerce, automated social media distribution, and specific marketing workflows tailored for brand owners and affiliate marketers.

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- Advantages:
 - Sora AI provides state-of-the-art, highly realistic text-to-video generation capabilities.
 - It can generate complex video sequences and maintain high visual fidelity based on detailed prompts.
- Disadvantages:
 - Sora AI functions solely as a generation model and does not provide an automated, campaign-based product library to manage inventory.
 - It lacks an integrated affiliate link management system and automated multi-platform distribution workflows.
 - Sora AI does not support a structured preview, approval, and systematic product image overlay processing pipeline required for commercial marketing.

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2.2.2.2 [Nano Banana](#) (Gemini)

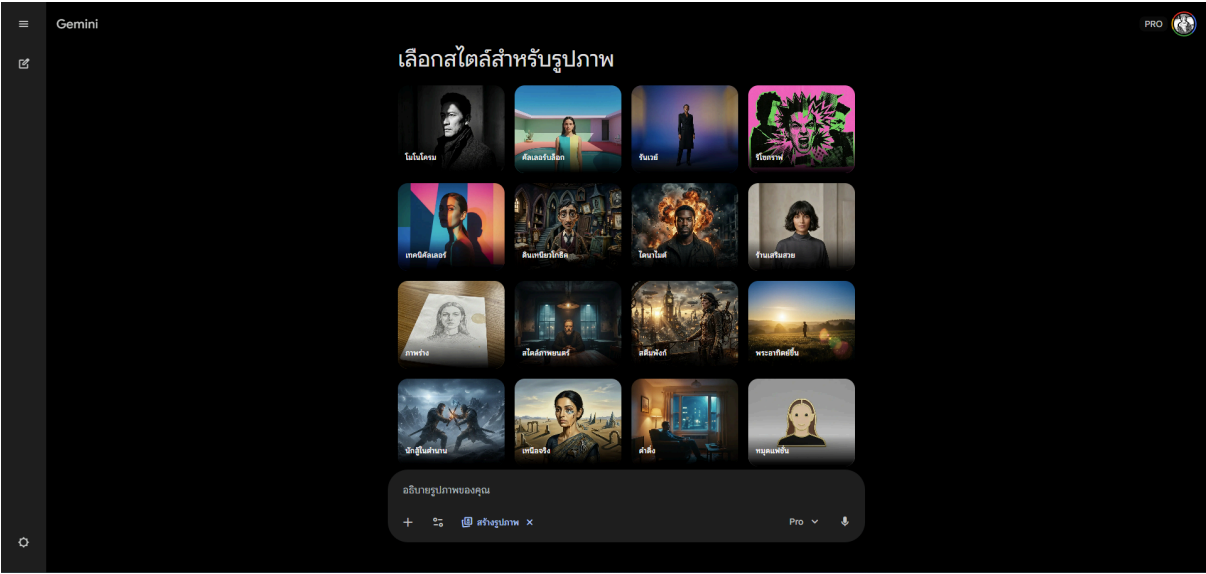


Figure 7: User interface of Nano Banana (Gemini), demonstrating its multimodal generative AI capabilities for image and text creation.

Nano Banana (<https://gemini.google.com/>) (powered by Google's Gemini models) represents a powerful suite of multimodal generative AI tools focused on text, image, and media generation. While it can produce high-quality visual content and analyze marketing copy effectively, it is not a dedicated social media management platform. Compared to ReelCast, it lacks the specialized infrastructure to connect generated media directly with product catalogs, automated affiliate systems, and API-based social media scheduling.

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- Advantages:
 - Nano Banana offers robust generative capabilities for analyzing selling points and creating platform-specific captions and hashtags.
 - It provides high-quality multimodal generation for images and text content.
- Disadvantages:
 - Nano Banana does not offer a dedicated, centralized product library system to manage campaigns and systematically track promotional performance.
 - It lacks built-in multi-platform distribution and automated affiliate link embedding functionalities.
 - It requires manual handling of media processing (such as FFmpeg overlays for brand logos and product images) and does not natively support an integrated commercial workflow.

In conclusion, after researching and comparing various AI tools and web applications—namely Predis.ai, InVideo AI, Sora AI (ChatGPT), and Nano Banana (Gemini)—against the Automated Multi-Platform Commercial Content Generation System (ReelCast), the comparative factors can be clearly distinguished as shown in the table below.

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Table 1 Tools/Software Comparison

Comparative Factors	Predis.ai	InVideo AI	Sora AI (ChatGPT)	Nano Banana (Gemini)	ReelCast
Video Generation	✓	✓	✓	✓	✓
Content & Scripting	✓	✓	✓	✓	✓
Campaign & Product Library	-	-	-	-	✓
Product Image Overlay	✓	-	-	-	✓
Multi-Platform Distribution	✓	-	-	-	✓
Affiliate Link Management	-	-	-	-	✓

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2.3 Technology Review

The Automated Multi-Platform Commercial Content Generation and Distribution System is built using a modern technology stack that encompasses frontend frameworks, backend services, advanced Generative AI models, and cloud infrastructure. The frontend is developed using React for the web browser interface and Flutter for the mobile application. The backend utilizes Python (FastAPI or Django) to handle complex AI processing. The core AI capabilities are powered by Google Cloud Vertex AI, specifically Google Veo for video generation and Gemini for text and multimodal tasks. FFmpeg is employed for media processing tasks such as scaling and image overlay. Data and files are managed using MongoDB and Google Cloud Storage, while the entire system is deployed on Google Cloud Run for a serverless infrastructure. More details about each component are provided below.

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2.3.1 Frontend Frameworks

The system utilizes a modern web technology to focus entirely on a responsive web application. This approach ensures an efficient Minimum Viable Product (MVP) delivery while providing a seamless user experience across both desktop and mobile devices.

2.3.1.1 Next.js



Figure 8: Logo of Next.js, the chosen frontend framework for building the ReelCast web application.

Technology Description: Next.js is a powerful, open-source React framework designed for building production-ready web applications. It extends React's capabilities by providing built-in features such as Server-Side Rendering (SSR), Static Site Generation (SSG), and an intuitive file-based routing system, resulting in highly optimized and fast-loading web pages.

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Alternatives Technology:

- Vue.js
- Standard React (with Vite or Create React App)
- PHP

The selection of this technology:

- Next.js provides Server-Side Rendering (SSR), which significantly improves the initial page load speed and overall performance of the platform's complex dashboards compared to a standard React Single-Page Application (SPA).
- It features a built-in routing system and API routes, streamlining the frontend architecture and simplifying data fetching from the Python backend.
- Because it is built on top of React, it retains all the benefits of a reusable component-based architecture while providing a robust, scalable environment for building a fully responsive web interface without the overhead of maintaining a separate mobile application codebase.

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2.3.2 Backend Framework

The backend system is responsible for handling AI processing, API integrations, and business logic.

2.3.2.1 Python



Figure 9: Logo of Python, the core programming language used for the backend layer.

Technology Description: Python is a high-level, versatile, and widely used programming language known for its readability and extensive ecosystem of libraries. In this project, Python serves as the core programming language for the Backend layer. It provides the foundation for the FastAPI framework, handles all business logic, orchestrates the AI generation pipeline (connecting to Google Veo and Gemini via Vertex AI), and manages media processing commands through FFmpeg.

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Alternatives Technology:

- JavaScript/TypeScript (Node.js)
- Java
- Go (Golang)

The selection of this technology:

- Provides the most robust official SDKs for Vertex AI, ensuring seamless integration with Google Veo and Gemini models.
- Offers efficient wrapper libraries (e.g., `ffmpeg-python`) to simplify complex video editing and brand overlay tasks.
- Natively supports `asyncio`, allowing the backend to handle long-running video generation tasks efficiently without blocking.
- Clean syntax and a vast ecosystem accelerate the delivery of complex APIs within the project's 6-month timeframe.

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2.3.2.2 FastAPI



Figure 10: Logo of FastAPI, the high-performance framework selected for backend API development.

Technology Description: FastAPI is a modern, fast (high-performance) web framework for building APIs with Python based on standard Python type hints. In this project, it is utilized as the core backend framework to construct the API endpoints, orchestrate the AI generation pipeline, and manage communication between the frontend and Google Cloud services.

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Alternatives Technology:

- Django (Python)
- Flask (Python)
- SpringBoot (Java)

The selection of this technology:

- FastAPI natively supports asynchronous programming ([async/await](#)), making it highly efficient for managing long-running, non-blocking tasks like waiting for Google Veo to generate videos or Gemini to produce text.
- Since it is a Python-based framework, it seamlessly integrates with the official Google Cloud SDKs (Vertex AI) and media processing wrappers like [ffmpeg-python](#).
- It provides exceptional speed and performance on par with NodeJS and Go, which is crucial for handling complex media processing workflows smoothly.
- It automatically generates interactive API documentation (Swagger UI), significantly accelerating the integration process between the backend and the frontend (Next.js) team.

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2.3.3 Generative AI Models

The core automated content creation of this project relies heavily on state-of-the-art Generative AI models accessed via Google Cloud Vertex AI.

2.3.3.1 Google Veo



Figure 11: Logo of Google Veo 3, the advanced generative video model utilized for B-roll generation.

Technology Description: Google Veo is an advanced generative video model capable of creating high-quality video content. In this system, it is specifically used to automatically generate marketing-focused background videos (B-Roll) or atmospheric footage based on the context of the product being promoted.

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Alternatives Technology:

- OpenAI Sora
- Luma Dream Machine
- Runway Alpha

The selection of this technology:

- Google Veo provides exceptional video fidelity and adherence to prompts, which is crucial for creating professional commercial Reels that pass the user's Preview & Approval workflow.
- It operates seamlessly within the Google Cloud ecosystem (Vertex AI), allowing for secure, scalable, and centralized API management.
- It significantly reduces the time and cost associated with traditional video production and stock footage sourcing.

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2.3.3.2 Google Gemini



Figure 12: Logo of Google Gemini, the multimodal LLM used for content analysis and scripting.

Technology Description: Google Gemini is a highly capable multimodal large language model (LLM) developed by Google. In this project, it is utilized to analyze product selling points from images or text and autonomously create optimized captions and hashtags specifically tailored for each target social media platform.

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Alternatives Technology:

- OpenAI GPT
- Kimi AI
- Anthropic Claude Sonnet

The selection of this technology:

- Google Gemini excels in multimodal tasks, meaning it can process and understand product images alongside text links accurately.
- It utilizes pre-defined marketing prompt templates within the system to ensure the generated captions and storyboards immediately capture audience attention without requiring prompt engineering skills from the user.
- Its advanced reasoning capabilities ensure that the generated marketing copy is engaging and contextually appropriate for platforms like Instagram and Facebook.

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2.3.4 Media Processing

To finalize the video content before distribution, the system requires a robust media processing engine to combine the AI-generated assets with the user's actual product images.

2.3.4.1 FFmpeg



Figure 13: Logo of FFmpeg, the media processing engine for video overlays and watermarking.

Technology Description: FFmpeg is a complete, cross-platform, open-source command-line tool used to record, convert, and stream audio and video. In this project, it is utilized as the core media processing engine for tasks such as editing, scaling, and overlaying actual product images and brand logo watermarks onto the AI-generated background videos.

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Alternatives Technology:

- GStreamer
- libVLC
- OpenCV (specifically for video/image matrix manipulation)

The selection of this technology:

- FFmpeg is the industry standard for programmatic video and audio manipulation, offering unmatched speed and reliability.
- It easily integrates with a Python backend (via wrappers like ffmpeg-python), allowing the system to automatically composite the final Reel and apply necessary watermarks to mitigate AI hallucination issues.
- It supports a vast array of codecs, ensuring the output video format is perfectly optimized for platforms like Instagram and Facebook.

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2.3.5 Database & Storage

The system requires a flexible database to manage content metadata, user data, and system logs, alongside a highly scalable storage solution for handling large video files.

2.3.5.1 PostgreSQL



Figure 14: Logo of PostgreSQL, the relational database management system for storing structured data.

Technology Description: PostgreSQL is a highly advanced, powerful, open-source object-relational database management system (RDBMS). In this project, it is utilized to securely store and manage structured data, including user accounts, content metadata, system configurations, relational mapping of affiliate tracking links, and the campaign-based Product Library.

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Alternatives Technology:

- MongoDB (NoSQL)
- SupaBased
- MySQL

The selection of this technology:

- PostgreSQL provides strict ACID compliance, ensuring the highest level of data integrity and reliability, which is critical when managing user profiles and commercial affiliate links.
- It offers robust support for the **JSONB** data type, allowing the system to store semi-structured data (such as dynamic API responses and captions generated by Gemini) within a relational database structure.
- It provides robust relational capabilities necessary for organizing products, generated videos, and tracking metrics into unified marketing campaigns/projects.

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2.3.5.2 Cloudflare R2



Figure 15: Logo of Cloudflare R2, the zero-egress fee object storage used for managing large video files.

Technology Description: Cloudflare R2 is an S3-compatible object storage service that allows developers to store large amounts of unstructured data without egress (bandwidth) fees. In this project, it is utilized to securely and cost-effectively store the large video files generated by Google Veo and processed by FFmpeg.

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Alternatives Technology:

- Amazon S3 (Simple Storage Service)
- Microsoft Azure
- Google Cloud Storage

The selection of this technology:

- Cloudflare R2 charges zero egress fees, which significantly reduces the operational costs of the system. This is highly beneficial since video files (Reels) consume substantial bandwidth when being retrieved and distributed to multiple social media platforms.
- It is fully S3-compatible, meaning it can be easily integrated into the Python (FastAPI) backend using standard and well-supported libraries like [boto3](#) without requiring architectural changes.
- It leverages Cloudflare's global edge network, ensuring extremely fast and reliable media delivery and access during the content generation and distribution process.

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2.3.6 Deployment Tools

The system requires robust deployment tools to containerize the application modules for consistent execution, alongside a highly scalable cloud infrastructure to manage the computational demands of AI processing and video generation.

2.3.6.1 Docker

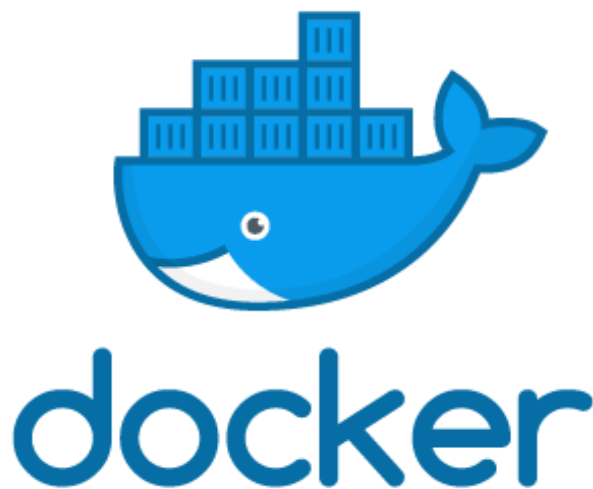


Figure 16: Logo of Docker, the containerization platform utilized for consistent system deployment.

Technology Description: Docker is an open-source platform that enables developers to build, deploy, run, update, and manage applications in isolated environments called containers. Containers package the application source code along with all its operating system libraries and dependencies required to run seamlessly in any environment. In this project, Docker is utilized to containerize the Frontend layer (Next.js), Backend layer (Python FastAPI), and Media Processing module (FFmpeg) to ensure operational consistency and simplify the deployment process on the AWS cloud infrastructure.

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Alternatives Technology:

- Podman
- Virtual Machines (e.g., VMware, Oracle VirtualBox)
- Native/Bare-metal Installation

The selection of this technology:

- Eliminates the "it works on my machine" issue by ensuring identical execution across local development and production environments.
- Prevents conflicts by securely separating complex system libraries (like FFmpeg) from backend packages (like FastAPI).
- Lightweight and fast-starting containers perfectly suit the project's Serverless deployment on AWS, enabling efficient resource scaling.

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2.3.7 Queue Management

Queue Management acts as a buffer that temporarily holds background tasks until they can be processed asynchronously. This prevents system overload and ensures the main application remains fast and responsive for users.

2.3.7.1 Redis



Figure 17: Logo of Redis, the in-memory data store functioning as the system's Queue Management.

Technology Description: Redis is an open-source, in-memory data structure store known for its high speed. In this project, it is utilized as a Queue Management system to handle asynchronous background tasks. It acts as a message broker, receiving tasks (Enqueue Task) from the backend layer (FastAPI) and distributing them (Dequeue Task) to the Worker Pool. This ensures that resource-intensive and time-consuming processes, such as video editing via FFmpeg and AI reel generation via Google Veo 3, are processed smoothly without blocking the main application's performance.

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Alternatives Technology:

- RabbitMQ
- Apache Kafka
- Amazon SQS (Since the infrastructure is hosted on AWS)

The selection of this technology:

- **High Performance:** Operating primarily in-memory, Redis provides exceptionally low latency and high throughput. This makes it highly efficient for managing fast-paced task queues between the backend and the worker pool.
- **Seamless Integration:** It integrates smoothly with Python-based environments (like FastAPI) and standard task queue libraries (such as Celery or RQ), ensuring stable and reliable asynchronous operations.
- **Lightweight and Scalable:** Redis is highly resource-efficient and runs flawlessly inside a Docker container, which perfectly aligns with the project's containerized infrastructure design.

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2.3.8 Infrastructure

To ensure the system is scalable, reliable, and easy to maintain without managing physical servers, a serverless deployment architecture is utilized.

2.3.8.1 AWS EC2



Figure 18: Logo of Amazon Web Services (AWS), the cloud infrastructure provider for deploying the system.

Technology Description: Amazon Elastic Compute Cloud (Amazon EC2) provides scalable computing capacity within the world's most comprehensive cloud platform, AWS. In this project, Amazon EC2 instances will be utilized to host and run the containerized frontend and Python backend services. This approach provides complete, granular control over the underlying virtual server infrastructure, allowing for custom configurations tailored specifically to the application's computing and networking needs.

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Alternatives Technology:

- Google Cloud Run
- Microsoft Azure Container Apps

The selection of this technology:

- AWS provides a highly mature, globally distributed infrastructure that ensures maximum uptime and reliability for the application.
- Utilizing Amazon EC2 offers deep control over compute resources and instance types. This allows the environment to be precisely tuned for heavy asynchronous workloads (such as managing concurrent connections while waiting for Google Veo and Gemini API responses). Coupled with EC2 Auto Scaling, the system can dynamically adjust capacity to handle traffic spikes while maintaining predictable performance.
- AWS offers unparalleled flexibility, robust Identity and Access Management (IAM), and seamless integration with other selected technologies like PostgreSQL and Cloudflare R2, making it an ideal environment for this project's deployment.

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Chapter 3 Quality Standard

3.1 ISO 29110 for Very Small Entity (VSE)

ISO 29110 is a guide applying to a very small entity, enterprise, organization, department, or project up to 25 people dedicated to software development. The guide provides project management and software implementation process which integrate practice based on the selection of ISO/IEC 12207 systems and software engineering software life cycle process and ISO/IEC 15289 software engineering software life cycle process guideline for the content of software life cycle process information product (documentation) standards elements.

3.1.1 Project management process

The purpose of the software management process is to establish and carry out in a systematic way the task of the software implementation project that allows complying with the project's objectives in the expected quality, time, and cost.

Selected process:

- 3.1.1.1 Project planning process
- 3.1.1.2 Project plan execution process
- 3.1.1.3 Project assessment and control process
- 3.1.1.4 Project closing process

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3.1.2 Software implementation process

The purpose of the software implementation process is the systematic performance of the analysis, design, construction, integration, and test activities for new or modified software products, according to the specified requirements.

Selected process:

- 3.1.2.1 Software implementation process
- 3.1.2.2 Software requirement analysis process
- 3.1.2.3 Software architectural design process
- 3.1.2.4 Software construction process
- 3.1.2.5 Software integration process and test process
- 3.1.2.6 Software delivery process

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Chapter 4 Project Plan

4.1 Motivation

To address the current bottlenecks in Social Commerce—where creators struggle with high-barrier video production and fragmented workflows across disjointed apps—this project aims to develop a unified workspace prototype. Because this is a Proof of Concept, the project objectives strictly avoid unmeasurable business goals such as sales revenue or engagement metrics. Instead, the focus is on concrete technical validation. The primary objective is to build a centralized platform integrating short video creation, affiliate link management, and mock scheduling to eliminate the "silo" effect. Success will be measured through User Acceptance Testing (UAT) to ensure functional stability, high task success rates, and a measurable reduction in time-on-task compared to traditional multi-app workflows. Furthermore, the system will be tested on its technical architecture, specifically its ability to navigate strict social media API limitations and execute proper error handling.

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4.2 Aims and Objectives

- **Develop an All-in-One Prototype:** Build the ReelCast system to centralize video creation, link management, and distribution, with success measured by the system's ability to pass core functional User Acceptance Testing (UAT).
- **Accelerate Production with AI Integration:** Integrate Generative AI (Google Veo and Gemini LLM) and validate the reduction in user task-completion time through usability testing, comparing the system's workflow against traditional multi-app processes.
- **Platform-Specific Content Generation:** Implement AI-driven features that successfully generate and format video and text outputs to meet the exact technical specifications and API requirements of target social media platforms.
- **Centralized Management Module:** Develop a functional product library that allows users to accurately perform CRUD (Create, Read, Update, Delete) operations for centralized affiliate link management.
- **Workflow Validation (Creative Control):** Implement a functional preview and approval interface, ensuring test users can successfully review, edit, and validate AI-generated content before the system processes the final export.
- **Automated Distribution & API Testing:** Implement automated publishing workflows to target social networks, measuring the system's success rate in transmitting data payloads and its ability to properly handle API rate limits and connection errors.
- **Comprehensive Technical Documentation:** Compile detailed system documentation, encompassing architecture diagrams, API integration references, and user guides, to ensure the long-term maintainability of the ReelCast codebase and clearly outline the system's structural logic for future development.

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4.3 Deliverables and Limits

4.3.1 Deliverables

4.3.1.1 System Architecture

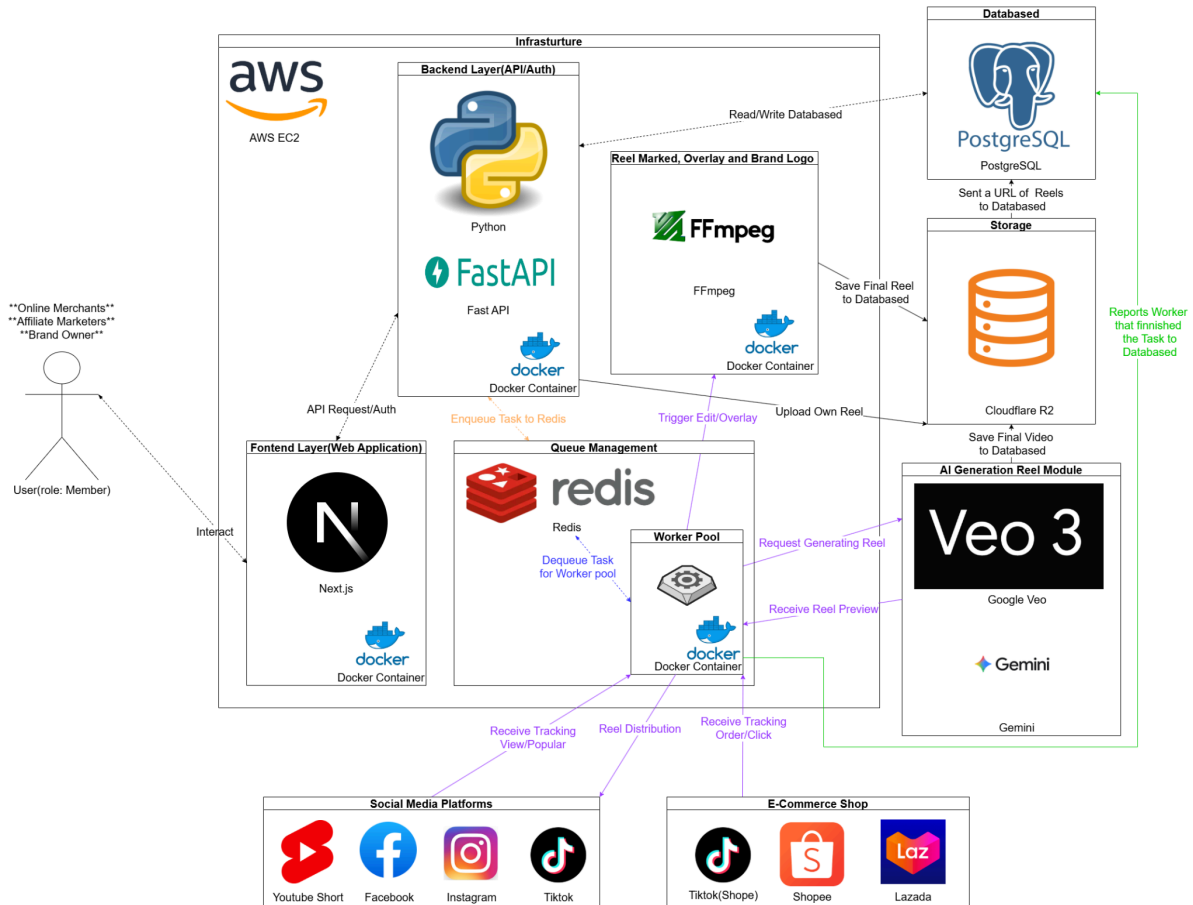


Figure 19: System Architecture diagram of the ReelCast platform, illustrating the interaction between Frontend, Backend, AI Module, Queue Management, and external integrations.

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The system architecture, hosted entirely on **AWS EC2** infrastructure, supports three main types of users: Online Merchants, Affiliate Marketers, and Brand Owners. The system's workflow and core components are described as follows:

- **Frontend Layer (Web Application):** Users interact with the system through a web application developed using Next.js running in a Docker Container. This layer handles the user interface, sends API requests for authentication and operations to the backend, and allows users to directly upload their own reels to Cloudflare R2 storage.
- **Backend Layer (API/Auth):** The core API layer is developed with Python and FastAPI, also containerized with Docker. It handles incoming user requests, manages direct read/write operations with the PostgreSQL database, and asynchronously enqueues processing tasks to Redis.
- **Queue Management (Redis & Worker Pool):** This acts as the asynchronous orchestrator for heavy processing. Redis queues the tasks, and the Worker Pool (running in a Docker Container) dequeues them to manage external executions. The Worker Pool handles:
 - Triggering the FFmpeg engine for video editing and overlays.
 - Requesting reel generation from the AI Module and receiving previews back.
 - Distributing finalized reels to social media platforms.
 - Receiving performance tracking data (views/popularity) and conversion metrics (orders/clicks).
 - Reporting task completion statuses directly to the PostgreSQL database.

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- **AI Generation Reel Module:** When triggered by the Worker Pool, this module utilizes **Google Veo 3** for high-fidelity video generation and **Gemini** for AI-driven content processing. It sends a reel preview back to the worker and saves the final generated video directly to Cloudflare R2 storage.
- **Real-time Processing (FFmpeg):** Triggered by the Worker Pool, this containerized FFmpeg module handles video finalization, including watermarking, overlays, and brand logo integration. The finalized reels are then saved to Cloudflare R2 storage.
- **Storage (Cloudflare R2):** Acts as the object storage for all large media files (user uploads, edited reels, AI-generated reels). Once a file is saved, it automatically sends the media URL to the PostgreSQL database.
- **Database (PostgreSQL):** Manages structural data, user information, file URLs, and task statuses reported by the Worker Pool.
- **External Integration:** The Worker Pool connects the system to two external ecosystems:
 - **Social Media Platforms:** Distributes reels to YouTube Shorts, Facebook, Instagram, and TikTok, while tracking views and popularity.
 - **E-Commerce Shops:** Connects with TikTok Shop, Shopee, and Lazada to track business conversions such as orders and clicks generated by the content.

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4.3.1.2 Document

- Proposal
- Project plan
- Software requirement specification
- Software design document
- Testing document
- Traceability record
- Software quality assurance document
- Video clips for demonstration
- A1 poster for presentation

4.2.1.3 Limits

- **Strict API Policies:** The system must navigate strict API limitations and frequently changing policies across different social media platforms, which may necessitate the use of Sandbox environments or complex automation workarounds.
- **AI Output Variability:** Because the system relies heavily on cutting-edge Generative AI models like Google Veo and Gemini, the generated video and text outputs can sometimes be unpredictable, requiring users to manually review and potentially regenerate content before publishing.
- **Resource-Intensive Processing:** Video generation and real-time media manipulation using FFmpeg are computationally heavy tasks, meaning the system relies on asynchronous processing that could delay immediate output during peak usage times.

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- **External Infrastructure Dependency:** The platform's core functionalities are heavily dependent on third-party cloud services, such as AWS for serverless deployment and Cloudflare R2 for storing large video files, linking the system's operational stability directly to the uptime of these external providers.
- **Cross-Platform Tracking Constraints:** While the system connects to e-commerce shops like TikTok Shop, Shopee, and Lazada to track promotional performance, comprehensive order tracking may be limited by the distinct privacy policies and API data restrictions of each individual marketplace.
- **Niche Target Audience Focus:** The platform is explicitly tailored for online merchants, affiliate marketers, and brand owners managing product catalogs. Consequently, its specialized campaign-centric features may not be suitable or flexible enough for general vloggers or non-commercial content creators.

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4.4 Software Process

4.4.1 Software Process Descriptions

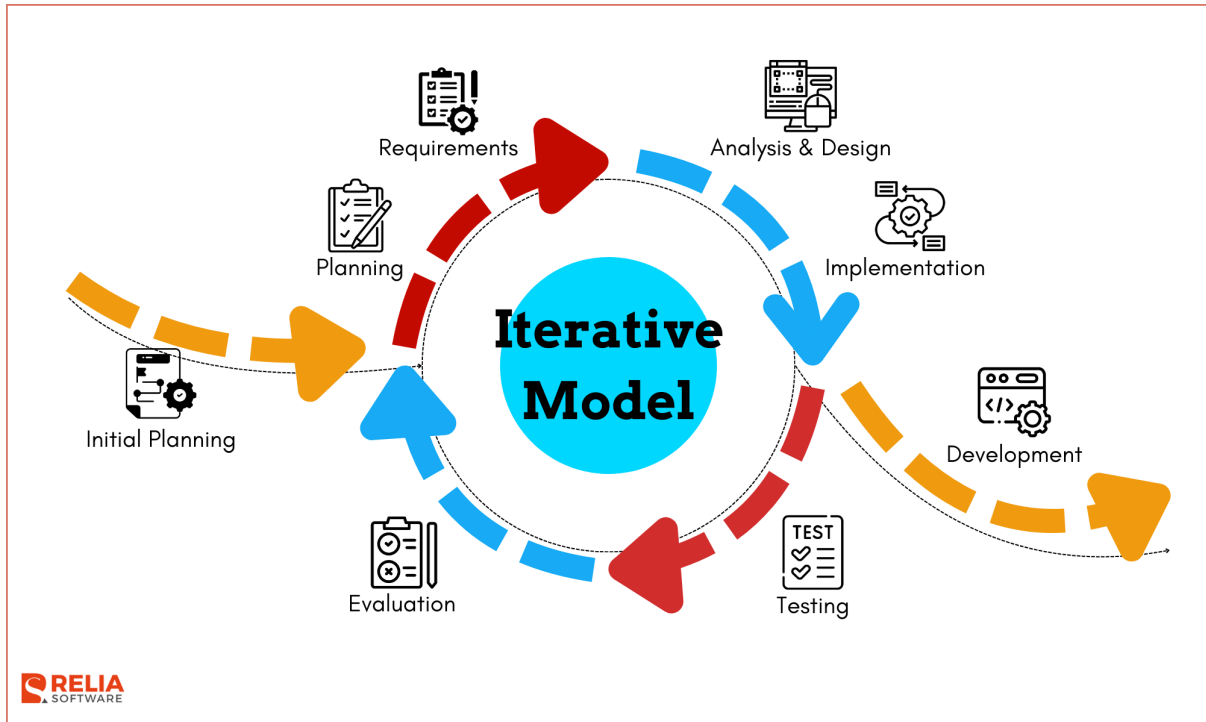


Figure 20: The Iterative Model software development lifecycle utilized for the ReelCast project.

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The Iterative Model is an approach to software development where the project is designed, implemented, and tested in repeated cycles (iterations). Instead of attempting to build the entire system at once, the development team starts with a simple, functional subset of the software and progressively enhances it with new features and improvements in each cycle until the final product is complete.

Key principles of the Iterative Model include:

- **Cyclic Refinement:** Building a basic, working version of a feature first, evaluating it, and expanding its capabilities through successive iterations.
- **Early Risk Mitigation:** Identifying and resolving technical challenges early by testing core functionalities and integrations in the initial development cycles.
- **Progressive Evaluation:** Reviewing the output at the end of each iteration to ensure the system architecture is stable before adding more complexity.

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4.4.2 How Iterative Relates to the ReelCast Project

Given the highly technical and experimental nature of ReelCast, the Iterative Model is essential. Here is how it specifically connects to your project's scope and constraints:

- **Handling Cutting-Edge AI Variability:** You are integrating advanced Generative AI models like Google Veo for video generation and Gemini for text analysis. Because AI outputs can be unpredictable, the Iterative Model allows you to build a basic generation pipeline in the first cycle. You can then evaluate the results, tweak your prompts, and refine the FastAPI backend in subsequent iterations rather than waiting for a finalized module within 2 November 2026
- **Perfecting the User Experience:** Your project requires a Preview & Approval system where users can review and regenerate videos if they are unsatisfactory. Using an iterative approach, you can start by developing a rudimentary prototype of this workflow, test the core data flow, and then iteratively refine the UI/UX to make it as smooth as possible for brand owners and affiliate marketers.
- **Managing Third-Party API Risks:** You noted a specific constraint regarding Social Media APIs, mentioning that you might need to use Sandbox environments or automation due to API policy limitations. The Iterative Model mitigates this risk; you can build and test the core publishing logic using mock data or sandboxes in early cycles. Once the system is stable, you can confidently integrate the actual APIs in a later iteration.
- **Parallel Development for a Small Team:** You have two distinct roles: a Backend & AI Engineer and a Frontend & Integration Engineer. The Iterative Model supports parallel work by allowing Person 1 to build the foundational API for the Product Library while Person 2 simultaneously builds the basic frontend UI for it. They can integrate their work at the end of a cycle to ensure the database and frontend communicate flawlessly before moving on to advanced features.

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4.5 Feature and Schedule

4.5.1 Features

There are two types of user:

- Member : Is a User are already Authenticate
- Guest : Is a User are not Authenticate

Feature 1: Registration & Authentication System

Actor: Guest/Member

Description: Guests can register to access the system, and Members can Authenticate their identity to log in.

Detail:

1. Guest can register to become a Member, as the system is designed for one primary user role including Online Merchants, Affiliate Marketers, and Brand Owners.
2. Member can authenticate to access the application and communicate with the backend via API requests.

Feature 2: AI Video Generation & Processing

Actor: Member

Description: Member can generate commercial background videos (B-roll) and platform-specific captions using AI

Detail:

1. Member can input a Prompt about Reels and select a product from product library
2. Member can automatically generate marketing-focused atmospheric background videos (B-Roll) using Google Veo.
3. Member can autonomously create optimized captions and hashtags specifically tailored for each target social media platform using Gemini.

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4. Member can upload their own Reels directly into the system for real-time processing and customization.
5. Member can overlay actual product images and brand logos onto the AI-generated background videos using FFmpeg.
6. Member can utilize a preview and approval workflow to maintain full creative control before any content is published.
7. Member can regenerate content if the initial AI outputs are unpredictable or unsatisfactory before finalizing the post.

Feature 3: Multi-Platform Distribution

Actor: Member

Description: Member can schedule, publish, and distribute finalized videos to multiple social media platforms.

Detail:

1. Member can seamlessly publish the finalized Reels directly to target platforms.
2. Member can select a schedule to publish the finalized Reels to target platforms.
3. Member can distribute generated Reels to connected platforms such as YouTube Shorts, Facebook, Instagram, and TikTok.

Feature 4: Product Library

Actor: Member

Description: Member can manage product inventory, embed (affiliate links, product images, brand logo, properties description) about the product and categorize products into collections.

Detail:

1. Members can efficiently manage their inventory and updating comprehensive product data, including images, brand assets, and detailed specifications.

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2. The system allows members to embed affiliate links into each product profile to facilitate seamless affiliate marketing operations.
3. Members can categorize products into custom collections, enabling organized management based on specific marketing campaigns, themes, or project requirements.

Feature 5: Data Tracking

Actor: Member

Description: Enables members to track, aggregate, and visualize performance data from integrated e-commerce shops and social media platforms to optimize their marketing strategies and ROI.

Detail:

1. Members can connect with major e-commerce platforms (e.g., TikTok Shop, Shopee, and Lazada) to synchronize tracking data for orders and click-through rates (CTR) generated via embedded affiliate links.
2. Supports integration with short-form video platforms (e.g., YouTube Shorts, Facebook, Instagram, and TikTok) to track engagement metrics such as views and content popularity from published Reels/Shorts.
3. The system provides comprehensive analysis and interactive data visualizations (graphs/charts) by consolidating data from both e-commerce and social media sources, allowing members to gain actionable insights into their overall performance.

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4.5.2 Schedule Plan

Milestone	Task	Milestone Criteria	Planned date
1.	Proposal	• Topic defined • Proposal reviewed • Proposal submitted • Proposal presentation	30 March - 10 April (1) 27 April - 1 May (2)
2.	Progress I	• Software requirement specific • Project Planning • Design Document • Implement feature 1, 2, 4 • Test feature 1, 2, 4 • Review Feature 1, 2, 4 • 1st Progress presentation	22 June - 26 June (1) 13 July - 17 July (2)
3.	Progress II	• Software requirement specific • Project Planning • Design Document • Implement feature 3, 5 • Test feature 3, 5 • Review Feature 3, 5 • 2nd Progress presentation	17 August - 21 August (1) 31 August - 4 September (2)
4.	Show pro	• Show pro preparation	30 September
5.	Final Progress	• Final progress preparation • Final progress presentation	19 October - 2 November

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4.5.3 Milestone

- Inception Phase (Proposal)

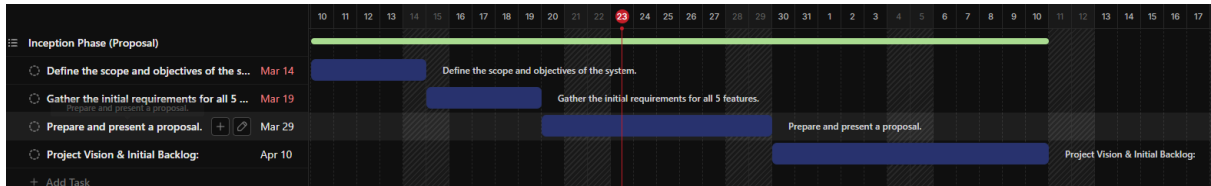


Figure 21: Project timeline and task scheduling for the Inception Phase (Proposal).

- Elaboration & Construction Phase 1 (Progress 1) (Iterative 1 Foundation)

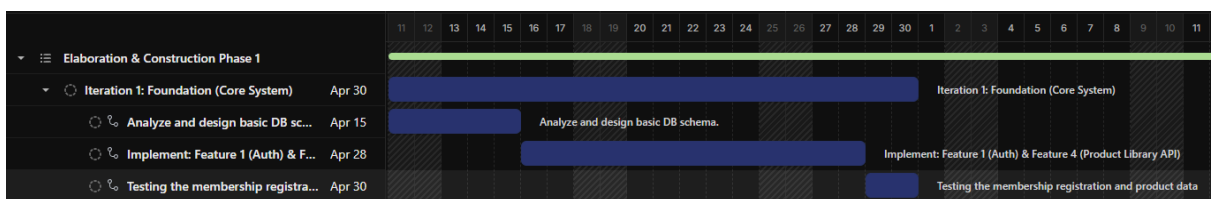


Figure 22: Project timeline for Elaboration & Construction Phase 1 (Iteration 1: Foundation).

- Elaboration & Construction Phase 1(Progress 1) (Iterative 2 AI Core)

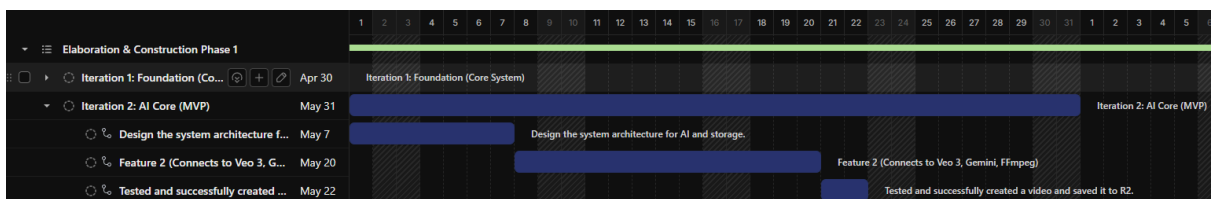


Figure 23: Project timeline for Elaboration & Construction Phase 1 (Iteration 2: AI Core MVP).

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- Elaboration & Construction Phase 1 (Prsentation Progress 1)

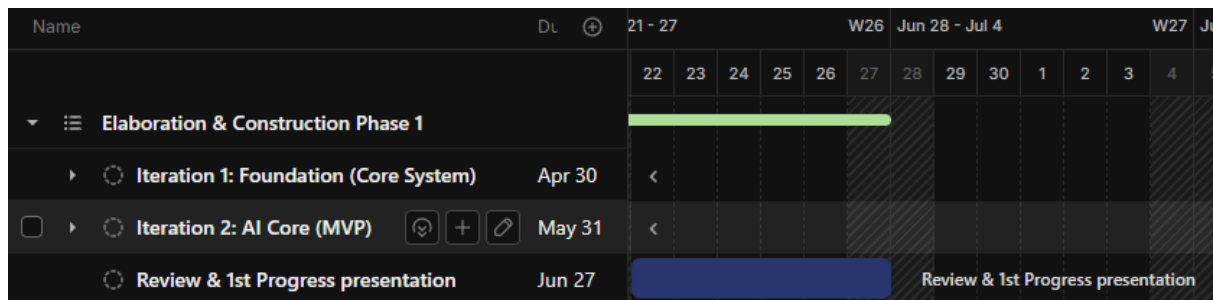


Figure 24: Project timeline for the 1st Progress review and presentation.

- Construction Phase 2 (Progress 2) (Iterative 3 Refinement & Distribution)

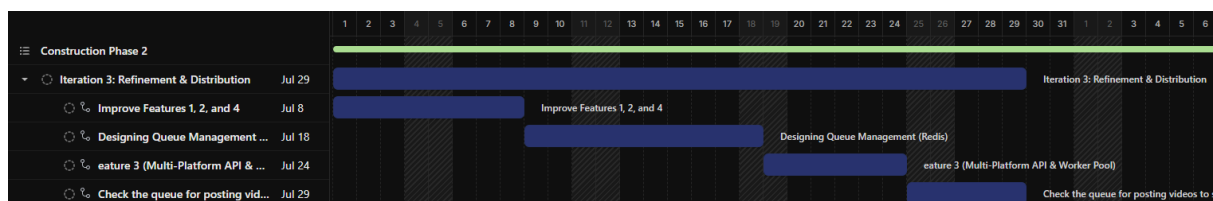


Figure 25: Project timeline for Construction Phase 2 (Iteration 3: Refinement & Distribution).

- Construction Phase 2 (Progress 2) (Iteration 4: Analytics)

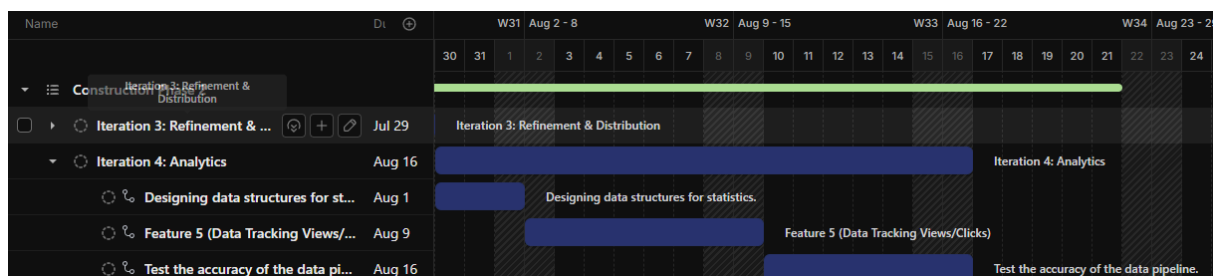


Figure 26: Project timeline for Construction Phase 2 (Iteration 4: Analytics).

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- Construction Phase 2 (Prsentation Progress 2)

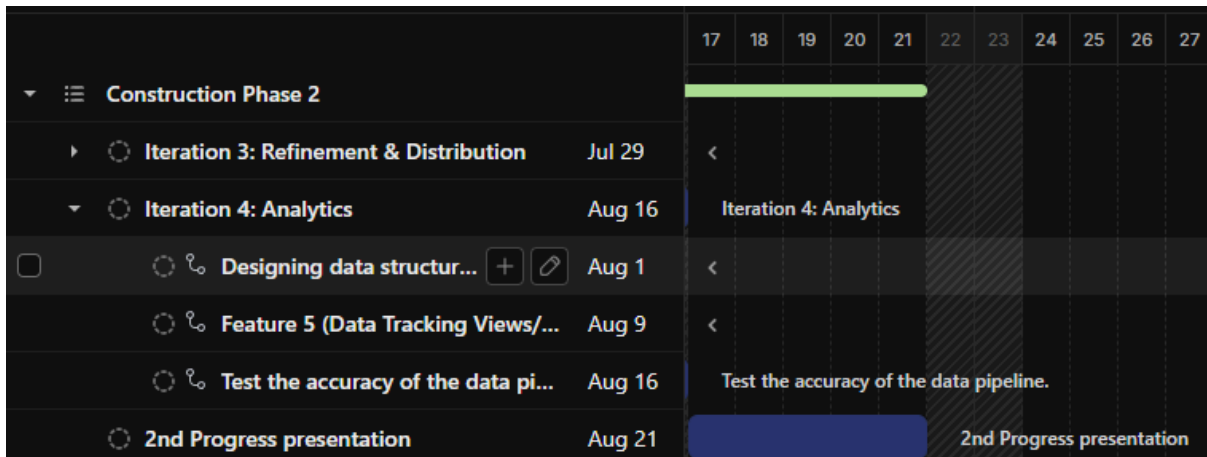


Figure 27: Project timeline for the 2nd Progress review and presentation.

- Transition Phase 1 (ShowPro) (Iteration 5: Integration & Hardening)

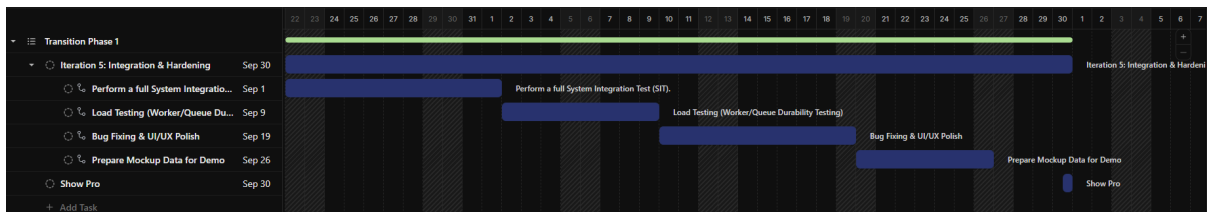


Figure 28: Project timeline for Transition Phase 1 (Iteration 5: System Integration Test and Bug Fixing).

- Transition Phase 2 (Final Progress) (Iteration 6: Final Release)

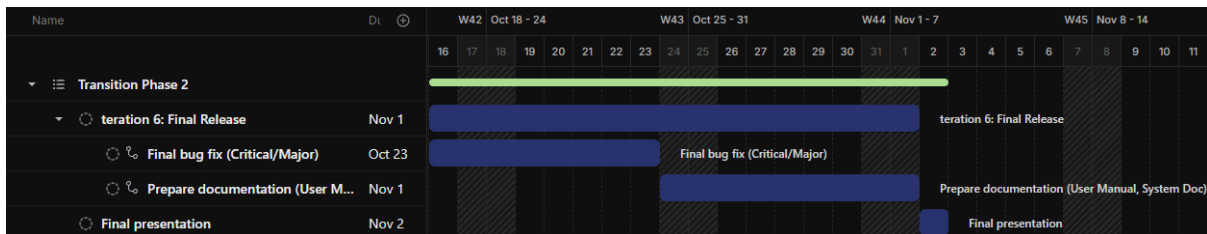


Figure 29: Project timeline for Transition Phase 2 (Iteration 6: Final Release and Documentation).

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